

# Combining Pattern Analysis, Threat Intelligence, and Domain Profiling for Smarter Phishing Detection

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**Abstract**—E-mail, spam, and phishing represent a significant threat to cybersecurity, exploiting both technological and human vulnerabilities. This study provides a thorough review of various approaches and models used to identify and mitigate these attacks. Research has consistently demonstrated machine learning (ML) and deep learning (DL) as key strategies for detection, using input vectors such as e-mail content, URLs, and website characteristics [3, 5]. Methods such as feature engineering, natural language processing (NLP), document enrollment, and topic modeling have contributed to improved classification accuracy [9].

Furthermore, improved identification rates have been achieved through hybrid and ensemble methods, including combinations of XGBoost, random forests, and logistic regression [4, 7]. Benchmark frameworks like PhishBench emphasize the importance of dataset diversity, model robustness, and real-time adaptability [2]. Several studies critique the limitations of traditional blacklists and heuristic systems, particularly in managing scalable data detection [6].

Deep learning architectures such as long short-term memory networks (LSTMs) and recurrent neural networks (RNNs) show promise in detecting evolving phishing tactics [1, 5]. However, challenges persist in generalization to unseen attacks, hyperparameter tuning, and the computational costs associated with extended training cycles [5]. The literature also highlights that phishing websites increasingly employ visual mimicry and HTTPS protocols, complicating detection further [6].

Open issues include the need for standardized evaluation protocols, more representative datasets, and the integration of behavioral signals [2, 5]. While ML and DL have significantly enhanced phishing detection, continuous advancement demands a holistic approach—combining technical innovation, robust datasets, and adaptive learning systems to counter rapidly evolving threats.

**Index Terms**—Phishing Detection, Machine Learning (ML), Natural Language Processing (NLP), WHOIS Domain Analysis, Random Forest, Feature Engineering, Cybersecurity.

## I. INTRODUCTION

In our increasingly connected digital world, phishing and spam emails present serious security risks. Misleading messages, malicious phishing, and spam emails pose major security concerns in our ever-linked digital society. Cybercriminals exploit user trust and behavior using deceptive messages and malicious websites, enabling identity theft, data leaks, and financial loss [6, 13]. Machine learning (ML) and deep learning (DL) have become essential tools for automated

detection, as traditional detection techniques cannot keep up with the sophistication of contemporary phishing tactics [1, 5]. This study investigates various methods for identifying spam and phishing using feature engineering, hybrid models, and sophisticated algorithms [4, 7]. The goals are improving accuracy, lowering false positives, and modifying to fit the always-shifting cybercrime tactics.

### A. Background

Phishing incidents pose as trustworthy sources and use social engineering to fool people into disclosing private information [13]. Attacks like these, which have become more sophisticated and frequently get past traditional filters, usually spread via malicious websites or emails [6]. Similarly, over half of all email traffic globally is spam, which exposes users to malware and decreases productivity. Research has moved from rule-based filters to ML/DL-based intelligent systems in the last ten years [3, 5]. These methods extract features from email content, URLs, or web elements, enabling automated classification of malicious content with high precision [9].

### B. Problem Statement

Because of their dynamic nature and capacity to evade traditional security measures, phishing and spam emails continue to be widely used despite advancements in detection technologies [2, 5]. Challenges include handling high-dimensional data, reducing false positives/negatives, and addressing the “curse of dimensionality.” There is a pressing need for scalable, accurate solutions that effectively integrate robust feature selection and extraction techniques to counter these threats [4, 5, 9].

- Many models lack adaptability to new and sophisticated attack patterns [1, 5].
- Feature extraction and selection processes can be complex and computationally expensive [4].
- Datasets are often imbalanced, outdated, or too narrow, impacting generalization and real-world performance [2].
- There’s no universal benchmarking framework to evaluate and compare detection methods consistently [2].

- To improve detection accuracy, develop hybrid models combining ML/DL algorithms with feature engineering [7, 9].
- Optimize dimensionality reduction techniques such as Principal Component Analysis (PCA) and Latent Semantic Analysis (LSA) for enhanced performance [4].
- Evaluate the efficacy of ensemble learning methods like Random Forest and XGBoost in phishing/spam classification [5, 7].
- Address computational efficiency while maintaining high precision and recall metrics [1, 5].
- Explore future research directions for scalable and adaptive cybersecurity solutions [5].

As phishing techniques evolve, there is a pressing need for scalable, robust, and real-time detection systems capable of identifying both known and zero-day threats [6, 14].

### C. Objectives

Despite significant progress, existing phishing and spam detection systems face several challenges and this collective research aims to:

- Many models lack adaptability to new and sophisticated attack patterns [1, 5].
- Feature extraction and selection processes can be complex and computationally expensive [4, 9].
- Datasets are often imbalanced, outdated, or too narrow, impacting generalization and real-world performance [2, 3].
- There's no universal benchmarking framework to evaluate and compare detection methods consistently [2].
- To improve detection accuracy, develop hybrid models combining ML/DL algorithms with feature engineering [7, 9].
- Optimize dimensionality reduction techniques such as Principal Component Analysis (PCA) and Latent Semantic Analysis (LSA) for enhanced performance [4].
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- Address computational efficiency while maintaining high precision and recall metrics [1, 5].
- Explore future research directions for scalable and adaptive cybersecurity solutions [5, 14].

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## II. LITERATURE REVIEW

### A. Introduction to the Domain

Significant cybersecurity threats include phishing emails and spam attempts to use digital channels to steal personal information. Utilizing machine learning (ML), deep learning (DL), and feature engineering has been the emphasis of recent

studies to improve the precision of detection and adaptability [4, 5]. This review synthesizes findings from 10 studies, emphasizing methodologies, performance outcomes, and unresolved challenges in phishing websites and email detection. Key themes include the efficacy of ensemble learning, NLP-driven feature engineering, and hybrid models, alongside gaps such as real-time scalability and evolving attack tactics [7, 9]. Email and the web have increased phishing and spam attacks, which pose significant threats to individuals, organizations, and national infrastructures [6]. As traditional detection systems struggle with new and adaptive tactics, researchers have increasingly focused on machine learning (ML), deep learning (DL), and natural language processing (NLP) for effective detection [4, 5]. The selected studies collectively examine various strategies, from email text classification and URL analysis to phishing website detection, highlighting innovations and ongoing challenges in the domain [1].

### B. Feature Engineering in Text-Based Email Detection

This literature review highlights key findings, trends, and essential research gaps, which look at the state of phishing detection research today. Developing strong cybersecurity defenses requires understanding the efficacy of current detection techniques and areas that need more research as phishing attacks increase in sophistication and volume. Gualberto et al. (2020), in "The Answer Is in the Text," proposed a multi-stage phishing detection method that leverages advanced feature engineering from email text [9]. Their pipeline integrates NLP preprocessing (lemmatization, vector space models), dimensionality reduction (Chi-Square, PCA, LSA), and ensemble learning (XGBoost, Random Forest). Notably, they achieved 100% F1-score on benchmark datasets, demonstrating that well-engineered features from email bodies alone can yield high detection accuracy.

#### 1) Prevalence and Impact

Phishing attacks have reached unprecedented levels in recent years. In 2023 alone, nearly 5 million phishing attacks were recorded, making it the worst year for phishing on record. The financial impact has been equally staggering, with business email compromise (BEC) costing victims worldwide more than \$50 billion according to FBI data. Organizations are increasingly vulnerable, with 84% experiencing at least one successful email-based phishing attack in 2022, resulting in direct financial losses increasing by 76% compared to the previous year.

#### 2) Evolution of Attack Vectors

While email remains the primary vector for phishing attacks, with an estimated 90% of successful cyber attacks starting with email phishing, attackers have significantly diversified their approaches. Mobile phishing has emerged as a growing threat, with 39% of mobile threats consisting of SMS phishing (Smishing). This shift toward multi-channel attacks represents a significant evolution in the phishing landscape [6].

Text-based business email compromise (BEC) attacks now represent 68% of all phishing attempts, indicating a

move away from attachment-based attacks toward more sophisticated social engineering tactics [13]. Furthermore, voice phishing (vishing), has significantly risen with incidents increasing by 260% year-over-year in Q4 2023.

### 3) Target Shifts

Furthermore, phishing attacks targeting strategies are dramatic. Social media platforms have become primary targets, comprising 42.8% of all phishing attacks in Q4 2023, a dramatic increase from 18.9% in Q3. Simultaneously, attacks against financial institutions decreased from 24.9% to 14% during the same period, suggesting attackers are adapting their strategies to exploit vulnerabilities in emerging platforms [6].

### 4) Impact of AI and Generative Technologies

Intelligent artificial intelligence technologies have greatly accelerated the frequency of phishing attacks, and generative AI technologies have dramatically accelerated phishing volumes. Since the launch of ChatGPT in 2022, phishing attacks have increased by an astonishing 4,151%. This correlation suggests that AI-powered tools have lowered the barriers to creating sophisticated phishing campaigns, enabling attackers to generate convincing content at scale [14].

### 5) Traditional Detection Methods

Blacklisting, rule-based systems, and signature-based techniques have been within the traditional forms of phishing identification. However, these methods have significant limitations. Blacklisting obstacles with a phishing link having a brief lifespan and the rapid emergence of new ones. While more effective for detecting zero-day attacks than blacklisting, signature-based approaches suffer from high false-positive rates. As phishing attacks become complex, these traditional approaches have proven increasingly inadequate [2, 6].

### 6) Machine Learning and Deep Approaches

Recent research shows a substantial shift toward machine learning (ML) and deep learning (DL) approaches for phishing detection. A review of 41 research studies from 2019 to 2024 reveals diverse approaches using feature engineering techniques such as character-level representations, word embeddings, and handcrafted features, combined with various model architectures, including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs/LSTMs), and hybrid models [1, 5, 9].

### 7) Feature Engineering Innovation

The importance of practical feature engineering is highlighted in several studies. One notable approach proposed a multi-stage method using Chi-Square and Mutual Information for dimensionality reduction alongside Principal Component Analysis (PCA) and Latent Semantic Analysis (LSA). This method achieved a 100% F1-measure using XGBoost and Random Forest algorithms with a significantly reduced feature set, demonstrating that sophisticated feature engineering can dramatically improve detection performance while reducing computational requirements [4, 9].

Another innovative study introduced an Optimal Feature Vectorization Algorithm (OFVA) that extracted 41 optimal intra-URL features (including 10 novel ones) from a large dataset of 274,446 instances. When combined with supervised machine learning algorithms, this approach developed a lightweight Random Forest model achieving 97.52% accuracy, further confirming the critical role of feature engineering in effective phishing detection [7].

### 8) Multi-Stage Detection Systems

Multi-stage detection methods combining several technologies get more and more attention in research. Natural language processing (NLP) combined with machine learning classification has shown promise. Usually including text pre-processing, feature extraction, dimensionality reduction, and classification, these methods have also demonstrated better performance than single-method detection systems by using the strengths of many techniques at every level [4, 9].

### 9) Demographic Factors

Studies have pointed out several demographic elements affecting phishing sensitivity. Gender has a part; men are more vulnerable to mobile phishing attempts than women. Another essential consideration is age; younger users often are more sensitive than older ones. In addition, affecting sensitive educational backgrounds, engineering and IT majors show decreased click rates on phishing emails compared to other fields [13].

### 10) Behavioral and Psychological Factors

Beyond statistics, phishing sensitivity is much influenced by psychological and behavioral aspects. Risky cybersecurity behavior is predicted by internet addiction and attentional and motor impulsivity. On the other hand, heavy computer use is associated with improved phishing identification capacity, implying that technological knowledge can be a preventive element. Studies reveal that behavior-based training can drastically reduce phishing events; staff members can enhance their phishing detection ability six-fold after six months. Still critical are user awareness and education [13].

### 11) Technical Challenges

Many technical problems still exist, even with significant progress. Zero-hour/zero-day attacks—which take advantage of unknown vulnerabilities—cause great difficulty for current detection systems [14]. Many detection techniques still suffer from high false-positive rates, which might cause alert fatigue and lower user confidence in security systems [5]. Additionally, there is limited research on robustness against adversarial attacks, where attackers deliberately design phishing attempts to evade detection systems. The real-time detection and integration challenge with existing security infrastructure remains insufficiently addressed [2, 5].

### 12) Data and Evaluation Limitations

Many studies rely on small or outdated datasets, which are limiting. Their generalizability to current phishing tactics. A notable lack of standardized evaluation bench-

marks across studies makes direct comparisons difficult [2]. Furthermore, insufficient diversity in datasets regarding geographical and organizational contexts limits our understanding of how phishing tactics vary across different environments [3].

#### 13) Emerging Threat Gap

Despite the dramatic increase in such attacks since the introduction of generative AI tools, there is limited research on detecting AI-generated phishing content. Multi-channel attacks that combine email, SMS, voice, and other communication methods are inadequately addressed in current research. Additionally, more studies are needed on countering the exploitation of trusted services for phishing purposes and defending against sophisticated multi-touch phishing campaigns that engage victims over extended periods [6, 14].

#### 14) Human-Technology Interface

The interaction between human factors and technological defenses remains underexplored. While we have some understanding of demographic and psychological factors affecting susceptibility, there is limited research on how to effectively personalize phishing training and detection systems based on these factors [13]. The impact of organizational security culture on phishing susceptibility also requires further investigation.

### C. Spam Email Detection Review and Taxonomy

The study by Tusher et al. (2024) categorizes phishing detection methods into various approaches, such as list-based, heuristic-based, visual similarity, machine learning (ML), deep learning (DL), and hybrid methods [8]. It is underlined as a potential solution because DL can automatically extract features from unprocessed data without human involvement. DL techniques are classified into discriminative (e.g., CNN, MLP), generative (e.g., AE, RBM), hybrid, ensemble, and reinforcement learning models. Ensemble DL models, which combine multiple algorithms, show higher accuracy than individual models like CNN and LSTM, which excel in handling multi-dimensional data and time-series sequences [1, 5].

CNN is efficient for feature extraction from raw data, while LSTM addresses long-term dependencies in sequential data. DL algorithms like CNN and LSTM excel in handling multi-dimensional data and time-series sequences, respectively. CNN is efficient for feature extraction from raw data, while LSTM addresses long-term dependencies in sequential data [1].

DL is appropriate for identifying zero-day phishing attacks since it removes manual feature engineering and can fit into big datasets [5]. Tests using several DL models found that although ensemble models—such as GRU-GRU—perform better in accuracy than single algorithms—they need longer training. This emphasizes the compromise between computational economy and accuracy. Metrics such as FPR, FNR, precision, recall, and AUC were used to evaluate model performance. Ensemble models consistently achieved higher accuracy rates [5, 7].

### Gap Identified:

While DL is promising, most existing implementations are limited by high computational cost, interpretability issues, and the lack of labeled, updated datasets representing current attack trends [2, 5].

- While DL automates feature extraction, traditional ML approaches still rely on manual feature engineering, which is labor-intensive and prone to errors [4, 9].
- Despite advancements in DL, detecting zero-day phishing attacks remains challenging due to rapidly evolving attack patterns [6, 14].
- DL models require significant computational resources for training and deployment. The increasing complexity of architectures exacerbates this issue [5].
- Many phishing datasets are outdated or lack diversity, leading to poor generalization in real-world scenarios. Additionally, labeled datasets are scarce due to privacy concerns [2, 3].
- Current DL models struggle with scalability and real-time detection due to high computational demands and the short lifespan of phishing attacks [6].
- Popular datasets for phishing detection include PhishTank (for URLs) and SpamAssassin (for emails). However, issues such as outdated or imbalanced datasets affect the robustness of DL models [2].

### D. URL-Based Phishing Detection Models

Karim et al. (2023) developed a hybrid phishing detection system using URL features. Their ensemble model (Logistic Regression + SVM + Decision Tree) combined hard/soft voting with feature selection and hyperparameter optimization. The approach was validated on over 11,000 URLs and showed superior performance across multiple metrics [7].

Phishing is still a significant cybersecurity risk since attackers constantly change strategies to surpass conventional detection systems. Particularly by considering URL characteristics and hybrid model architectures, recent research has concentrated on using machine learning (ML) and deep learning (DL) to improve phishing detection [4, 5, 7]. This review synthesizes the key results using current empirical research and thorough reviews and points out ongoing field shortcomings.

Machine learning models, primarily ensemble and hybrid classifiers, consistently outperform traditional list-based and rule-based systems in detecting phishing websites [6]. Notably, random forest (RF), decision tree (DT), support vector machine (SVM), and hybrid models combining these algorithms have achieved high accuracy rates [7].

The effectiveness of phishing detection models depends on the quality and relevance of features extracted from URLs. Recent studies have introduced novel intra-URL features and optimal feature selection algorithms (e.g., OFVA, canopy selection), significantly improving detection performance [4, 7].

Combining multiple classifiers through bagging and boosting (e.g., hybrid models of SVM, DT, RF, XGBoost) yields superior results compared to single-model approaches. These hybrid systems leverage the strengths of individual algorithms,

achieving accuracy and outperforming state-of-the-art techniques on standard datasets [7].

Using extensive, diverse datasets (e.g., over 270,000 URLs) enables models to generalize better and detect a broader range of phishing attacks, including zero-day threats [7, 14]. Studies emphasize the importance of dataset size and diversity for robust model training and evaluation [2].

Comprehensive evaluation using accuracy, precision, recall, specificity, and F1-score metrics is standard practice. Ensemble models, particularly RF and hybrid classifiers, consistently achieve high scores across these metrics, indicating balanced performance in detecting phishing and minimizing false positives [5, 7].

#### *Gap Identified:*

- The study is URL-centric, meaning it cannot detect phishing attempts that do not rely on malicious links (e.g., text-only fraud). Also, the dataset's representativeness is limited to known phishing domains [6].
- Traditional detection methods and some ML models remain static, struggling to adapt to the rapidly changing tactics of phishers. This results in high false-positive and false-negative rates, especially for zero-hour attacks. There is a need for more adaptive, continuously learning systems that can evolve alongside phishing techniques [14].
- While novel feature engineering has improved detection, there is still debate over which features are most predictive and how to justify their inclusion. Some models use many features without a clear rationale, potentially introducing noise and reducing interpretability [4].
- Most research is conducted on benchmark datasets, with limited validation in real-world, operational environments. The transferability of models across different languages, regions, and types of phishing attacks remains an open question [2, 3].

#### *E. Review of Email Classification Research Trends*

Email classification is critical in cybersecurity and information management, particularly for spam and phishing detection. The literature reveals a rich landscape of machine learning (ML) and features engineering approaches but also highlights persistent challenges and research gaps.

Mujtaba et al. (2017) offered a decade-long review of email classification trends. They categorized methods into spam, phishing, and multi-folder classification, highlighting the dominance of bag-of-words and TF-IDF feature engineering at the time. The paper identifies a gradual shift toward ML classifiers and the importance of proper preprocessing, feature selection, and dataset diversity [3].

The effectiveness of email classification models heavily depends on the selection and extraction of features. Commonly used features include those from the email header, body, URLs, JavaScript, and behavioral patterns. Recent research emphasizes the importance of combining multiple feature

types, including human behavior features, to improve classification accuracy [9].

Public datasets such as PU, SpamBase, Enron, and Phishing-Corpus are widely used for benchmarking. However, there is a notable scarcity of large, diverse, and up-to-date datasets for phishing email classification, which limits the generalizability of research findings [2].

Accuracy, precision, recall, F-measure, and false positive/negative rates are the standard metrics for assessing classifier performance. However, some studies highlight the need for additional metrics like area under the curve (AUC) and classification time, especially when dealing with imbalanced datasets or real-time applications [5].

While traditional ML models dominate, there is a growing interest in deep learning and hybrid approaches that can automatically learn feature representations and adapt to evolving threats. These methods show promise, particularly for handling complex and high-dimensional data [1, 5].

#### *Gap Identified:*

- The review predates the DL wave and does not cover neural networks or transformer-based models that now dominate the field [3].
- Most existing models are trained on static datasets and lack mechanisms for real-time adaptation. The challenge of concept drift—where the characteristics of spam and phishing emails evolve over time—remains largely unaddressed. Only a minority of studies have explored online or incremental learning approaches [5].
- Most available datasets are in English, and there is limited research on multilingual email classification. This language bias restricts the applicability of models in non-English contexts. The lack of standardized datasets for specific application areas, such as phishing, also introduces bias and hampers progress [2].
- Most research focuses on text-based features, yet spammers increasingly use images to bypass filters. Only a few studies have incorporated image analysis, indicating a gap in addressing image-based spam and phishing [6].

#### *F. Detection of Phishing Websites*

Still a significant risk in cyberspace, phishing uses psychological manipulation to fool people into revealing private information. Detection systems have developed from simple blocklist systems to advanced machine-learning models [5, 6].

Scholarly work stretching a broad spectrum reflects this development, looking at detection techniques from many technical angles. Zieni et al. (2023) polled approaches to phishing website detection, categorizing them as list-based, similarity-based, or ML-based. They underlined how obfuscation methods—e.g., SSL abuse and visual mimicry—have evolved to render conventional detection useless. Their thorough analysis showed a growing dependence on dynamic elements and browser-based signals [6].

One of the earliest and most straightforward strategies for detecting phishing websites involves maintaining blocklists

(databases of known malicious URLs) and allow lists (trusted sites). Browsers like Google Chrome and Mozilla Firefox integrate services like Google Safe Browsing to block harmful sites. While these methods are efficient and easy to implement, they fall short when confronting zero-day attacks—newly created phishing pages not yet listed [14].

To address this, researchers have explored heuristic and predictive methods to enhance list generation. For instance, techniques have been proposed involving URL pattern analysis, DNS lookup, and similarity matching with known phishing sites [5, 6]. Despite these enhancements, list-based approaches remain inherently reactive and are most effective when combined with other detection layers.

Another prominent class of detection mechanisms focuses on content similarity—comparing suspicious web pages with known legitimate ones. This can involve either textual content (HTML structure, Document Object Model (DOM), CSS) or visual features (layout snapshots, logo detection, and color schemes) [6].

Text-based techniques apply algorithms like TF-IDF, DOM-tree matching, or heuristic layout analysis. On the visual side, methods such as Scale-Invariant Feature Transform (SIFT), Earth Mover’s Distance (EMD), and Optical Character Recognition (OCR) are used to detect copied visual patterns or logos [6, 14].

Though robust against many obfuscation techniques, these approaches are resource-intensive in computation and storage. They also face limitations from evasive tactics used by attackers, including the replacement of text with images or intentional layout distortions [6].

Machine learning (ML) has emerged as the most dynamic and scalable solution for phishing detection. Here, the task is framed as a binary classification problem, where web pages are categorized as legitimate or phishing based on extracted features [5, 7].

Feature engineering is central to ML-based methods. Features are typically derived from:

- URL attributes (e.g., length, entropy, domain name structure)
- HTML and JavaScript content (e.g., login forms, anchor links)
- Traffic properties (e.g., HTTP headers, TLS certificate data)
- Reputation signals (e.g., page ranking, DNS queries)

Various models have been employed, ranging from classical classifiers such as Random Forest, Support Vector Machines (SVM), and Decision Trees to deep learning models like Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks [1, 4, 7]. Online learning algorithms, which update models incrementally, have also been introduced to improve adaptability to evolving phishing tactics [14].

While ML methods exhibit high accuracy and adaptability, their effectiveness is highly contingent on the quality and diversity of training datasets. Many studies rely on privately

curated datasets, often using sources like PhishTank and Alexa, limiting reproducibility and standardization across the field [2].

The literature points to growing interest in hybrid models, in which list-based filters combine machine learning or similarity analysis to improve detection accuracy and lower false positives. Moreover, ensemble learning methods combining several classifiers have shown encouraging performance optimization [4, 7].

#### *Gap Identified:*

The study highlights the lack of adaptive, real-time solutions that identify sophisticated zero-day phishing websites, even as it suggests some encouraging methods. Though much has been done, several issues still exist:

- Lack of standardized public datasets makes benchmarking difficult [2, 3].
- Real-time detection remains computationally expensive, especially with visual comparison methods [6].
- Feature selection and engineering remain critical bottlenecks in achieving scalable deployment [4].
- The explainability of machine learning models is under-explored, especially in high-stakes security applications [5].

#### *G. Deep Learning for Phishing Websites*

A significant cybersecurity issue is phishing websites, in which attackers always develop ways to pass for real websites and fool people into disclosing private information. Based on systems of advanced machine learning and deep learning models, the academic community has investigated many approaches to identify such risks in great detail [5].

Zara et al. (2024) proposed using DL models like LSTM and GRU, combined with ensemble learning, for phishing website detection [1]. Earlier phishing detection systems relied heavily on blocklists and allowlists, storing known malicious or trusted URLs. Though these techniques offered quick detection, they proved insufficient against zero-day attacks — novel phishing pages not yet cataloged [14].

Using relational rule mining and heuristic methods, including the CANTINA approach—which made use of TF-IDF scores and content features—many studies. Although these techniques increase interpretability, they are not easily flexible enough to fit changing phishing techniques and often call for manually selected rules [6].

Using image processing methods, including wavelet hashing, SIFT, and color histograms, another crucial field of research compares phishing websites with real ones. Although visually focused models can detect look-alike phishing pages, they are computationally expensive and may fail against sites that obfuscate visual cues or use dynamic content [6, 14].

ML-based methods have gained prominence by leveraging engineered features like URL structures, HTML content, domain metadata, and email headers. Algorithms such as Random Forest (RF), Support Vector Machines (SVM), and K-Nearest Neighbors (KNN) have shown promising results with

accuracies typically between 90–97%, depending on dataset quality and feature selection techniques [4, 7].

Feature engineering, often using Information Gain, Gain Ratio, or Principal Component Analysis (PCA), is critical in identifying impactful attributes like SSL final state [4, 9]. Recent studies have pivoted towards Deep Learning (DL) methods, including Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU). These models can learn complex temporal dependencies from sequential data like URLs or webpage structures [1].

Ensemble methods like AdaBoost, XGBoost, and Random Forest have been integrated with DL models to enhance prediction accuracy [7]. The current study demonstrated that Random Forest achieved a 99% accuracy, outperforming other contemporary models in phishing detection tasks [4].

#### *Gap Identified:*

Despite the advancements detailed above, several critical gaps and limitations remain in the literature:

- Most studies rely on dated or private datasets (e.g., from PhishTank or UCI), leading to reproducibility issues. There’s a need for continuously updated and publicly accessible datasets that reflect real-world phishing trends and threats [2].
- Although ensemble and deep learning models perform well on known patterns, their robustness against zero-day phishing attempts is underexplored. Future work should investigate adversarial training or GAN-based phishing simulations to improve model generalizability [14].
- DL and visual similarity models often require significant computational resources, making real-time deployment a challenge. Optimizing inference time and reducing memory footprint are essential for practical applications in browsers and firewalls [5, 6].
- While structural and content features dominate current approaches, user behavior patterns, network activity, and interaction timing are still underutilized. Incorporating such contextual data may offer stronger cues for identifying sophisticated phishing sites [13].
- Phishing schemes vary across languages and regions, yet few models are evaluated on non-English URLs or multilingual phishing content. This gap limits the effectiveness of global anti-phishing systems [3, 6].

#### *H. Benchmarking Framework for Phishing Research*

El Aassal et al. (2020) introduced PhishBench, a benchmarking framework for systematically comparing phishing detection techniques. They also proposed a taxonomy of features, highlighting the need for standardized evaluation environments and identifying weaknesses in current literature, such as imbalanced datasets and poor cross-study comparability [2]. Phishing detection research has advanced significantly, beginning with heuristic and rule-based methods and evolving into machine learning-driven frameworks [5, 6]. Traditional models relied on blacklists and manual feature extraction,

which were effective against known threats but often failed in the real-time detection of zero-day attacks [14].

With the rise of AI, researchers shifted toward supervised learning techniques. Researchers turned to supervised learning methods, including HTML content, URL structure, and email metadata. Methods, including Random Forest, Support Vector Machines, and Gradient Boosting, have been extensively embraced for phishing attempt classification depending on engineered features [4, 7]. However, comparative results across studies were sometimes tricky, given the variances in datasets, features, and evaluation criteria [2].

El Aassal et al. identify a key limitation in the previous literature: the absence of a unified benchmarking framework. Most studies evaluate their techniques under different experimental conditions, with non-standard datasets and varying classifiers, which undermines reproducibility and generalizability. Moreover, the performance of phishing detection models is often exaggerated due to the use of balanced datasets, which do not reflect real-world conditions where phishing emails are far less frequent than legitimate ones [2].

The authors introduce PhishBench, a customizable benchmarking platform incorporating over 200 phishing-related features, multiple classifiers, and standardized email, URLs, and website datasets to address this. This framework supports feature ranking, real-time evaluation, and imbalanced learning strategies, an innovation absent in prior research tools.

The study proposes a novel taxonomy of features categorized across three linguistic levels—syntactic, semantic, and pragmatic—and further segmented into lexical, network, and script attributes [2]. For instance:

- Syntactic features include structural patterns like TLD positioning in URLs.
- Semantic features capture interpretive meaning, such as urgent language in email subjects.
- Pragmatic features reflect contextual or behavioral signals, such as script behaviors and WHOIS data [18].

This taxonomy unifies feature classification across phishing vectors (emails, URLs, websites) and provides a common language for evaluating model inputs.

Through PhishBench, the study systematically evaluates classifier performance under both balanced and imbalanced scenarios. Key insights include: Classifiers like Random Forest and AutoML (e.g., AutoLearn) maintain high performance even under data imbalance [7]. Email header features significantly enhance detection accuracy. Feature selection using Information Gain and Recursive Feature Elimination consistently identifies a small set of highly informative attributes, reducing computational costs without significant performance loss [4].

Notably, many studies fail to explore feature extraction times, critical for real-time applications. The authors highlight this as a key oversight and provide detailed timing analyses [2].

Although AutoML tools such as TPOT and AutoSklearn have become popular, the paper underlines that most frameworks run just on preprocessed data and lack integrated feature

extraction. This limitation further complicates benchmarking across studies.

The few available tools, such as URLNet, focus on narrow use cases (e.g., neural networks on URLs) and don't offer extensibility or comparative evaluation capabilities. PhishBench fills this void with modular architecture combining feature engineering, categorization, and evaluation.

#### *Gap Identified:*

Although PhishBench is valuable, adoption in the broader research community remains limited, suggesting a need for wider awareness and integration into future phishing research. Despite the richness of phishing detection literature, the following key gaps remain:

- **Lack of a Universal Evaluation Framework.** Prior work lacks standardized platforms to compare feature sets and classifiers under uniform conditions [2].
- **Imbalanced Data Handling.** Many models are trained on balanced datasets, which do not reflect real-world distributions where phishing examples are sparse [5].
- **Feature Extraction and Timing.** Most research overlooks the latency introduced by complex features, limiting deployment in real-time settings [4].
- **Reproducibility Challenges.** The absence of shared codebases and datasets makes reproducing or validating experimental results difficult.
- **Underexplored Pragmatic Features.** Features related to attacker intent and contextual behavior are less studied despite their potential for robust detection [13].
- **Limited Integration of Online and Deep Learning Models.** Although ensemble and AutoML approaches show promise, their deployment in phishing detection is still evolving [1, 7].

#### *I. Feature Engineering with Topic Modeling*

Another paper by Gualberto et al. (2020), "From Feature Engineering to Enhanced Prediction Rates," extended earlier work by integrating topic modeling (LDA) into phishing detection [4]. Early phishing detection models largely depended on explicit features such as the presence of URLs, suspicious tags, IP addresses, or link frequency. These approaches often incorporated classifiers like Random Forest and Bayes Net and leveraged datasets such as PhishingCorpus and SpamAssassin [17].

To tackle the high-dimensional nature of text data, methods using vector space models such as TF-IDF and Bag of Words (BoW) became common [9]. These models numerically represent text but are often sparse and devoid of contextual semantics. Subsequent efforts addressed these limitations through dimensionality reduction techniques like Singular Value Decomposition (SVD) and Non-negative Matrix Factorization (NMF).

One of the most promising advances came from using Latent Dirichlet Allocation (LDA). This probabilistic model identifies latent topics within a corpus, enabling compact, low-dimensional representations of documents. Gualberto et al.

implemented LDA as a central part of their detection system, achieving high prediction accuracy by capturing deeper thematic structures of phishing versus legitimate emails [4].

Additionally, the study applied lemmatization enhanced by WordNet, allowing the semantic normalization of terms. This process improved generalization and reduced the vocabulary size without compromising performance, improving F1 scores and classification consistency.

The study benchmarked machine learning algorithms, including XGBoost, Random Forest, Decision Trees, Logistic Regression, SVM, and MLP. The proposed model achieved near-perfect performance (F1-score of 99.95) using XGBoost with features derived from LDA topics [4, 9].

This success is attributed to a rigorous preprocessing pipeline, semantic feature engineering, hyperparameter optimization, and excluding non-textual metadata, focusing solely on the email body.

Not too far behind were distributed representations, including Word2Vec, Doc2Vec, and FastText. These models capture syntactic and semantic relations by embedding words or whole documents in continuous vector spaces. Although efficient, these techniques usually need more computational resources and advanced architectures like CNNs or RNNs to reach comparable degrees of accuracy [1].

Some studies also introduced hybrid models combining NLP with behavioral features (e.g., user interaction patterns), but these were not the focus of Gualberto et al.'s work, which maintained emphasis on textual content only.

#### *Gap Identified:*

Despite the notable contributions in this domain, several key gaps remain:

- **Limited Dataset Diversity.** Most evaluations rely on the same datasets (e.g., PhishingCorpus, SpamAssassin), which may not represent modern phishing techniques, particularly those leveraging multimedia or advanced obfuscation [2, 17].
- **Over Reliance on Email Body Only.** While focusing on the body improves generalization, excluding header or link metadata may overlook valuable indicators in real-world phishing scenarios [5].
- **Computational Cost of Feature Engineering.** While LDA and semantic processing improve results, they introduce added computational complexity, potentially limiting real-time applications [4].
- **Limited Use of Behavioral Data.** Most models including the one in this paper, ignore behavioral signals such as click patterns or device identifiers, which could provide supplementary evidence of phishing [13].
- **Underexplored Ensemble Learning in Textual Contexts.** Although ensemble methods like XGBoost are used, integrating multiple semantic or contextual perspectives in ensemble architectures remains underutilized [4].

### *J. Deep Learning in Phishing Detection: Challenges and Future Direction*

A review of the systematic literature on DL in phishing identification by Do et al. (2022) provided a taxonomy highlighting difficulties, including long training time, manual parameter tuning, and limited explainability by grouping models by architectural and feature input type, along with future directions, few-shot learning, adversarial training, and real-time detection systems [5].

The study stands out by providing:

- A structured taxonomy of phishing attack vectors and DL techniques.
- A comparison of DL algorithms' strengths and limitations.
- An empirical evaluation of selected DL models to benchmark performance.
- Notably, the study highlights manual hyperparameter tuning, long training times, and inconsistent accuracy across datasets as pressing issues in DL-based phishing detection [5].

Do et al. conducted a systematic literature review (SLR) of 81 selected studies from 2018–2021 to evaluate the use of DL in phishing detection. They proposed a taxonomy that organizes detection techniques into categories based on media (e.g., websites, email, SMS, social networks), method (e.g., blacklist, heuristic, ML, DL, hybrid), and architecture [5].

Detection Medium:

- Phishing attacks target users through multiple platforms.
- Websites impersonating trusted sources.
- Emails using spear-phishing and whaling tactics.
- SMS (Smishing), VoIP (Vishing), and mobile apps.
- Social media platforms leveraging trust-based relationships [6, 13].

The authors identified six primary detection methods:

- List-Based: Uses blacklists or whitelists of known URLs [14].
- Heuristic-Based: Relies on manually crafted rules.
- Visual Similarity: Compares webpage snapshots for resemblance [6].
- Machine Learning: Trains models using features like domain age or link count [4, 7].
- Deep Learning: Leverages CNN, RNN, LSTM, etc., for automatic feature learning [1, 5].
- Hybrid: Combines multiple models to improve accuracy and robustness [7].

A range of deep learning models has been utilized for phishing detection:

- Convolutional Neural Networks (CNNs): Effective in image and layout recognition (e.g., screenshots of phishing pages).
- Recurrent Neural Networks (RNNs) and LSTM: Suitable for sequence-based data like URLs or text.
- Autoencoders and GRUs: Applied for dimensionality reduction and anomaly detection.

- Multi-Layer Perceptrons (MLPs): Often used for structured tabular data classification [1, 5].

DL models outperform conventional methods, particularly in training on big, varied datasets. For best results, these models do, however, often need major tuning and are resource-intensive [5].

*Gap Identified:*

- The study rightly calls out the lack of standardized datasets and real-world simulations. Most DL models are tested in isolated lab conditions, limiting their deployment potential [2, 5].
- Many studies use synthetic or outdated datasets. There's a need for dynamic, real-time datasets reflecting modern attack strategies [3, 17].
- Current DL models struggle with unknown phishing patterns. Enhancing models' ability to generalize to new threats remains an open issue [5, 14].
- DL methods often require high-performance computing resources, which limits their deployment in resource-constrained environments like mobile apps or browsers [1].
- Most DL systems function as "black boxes." Efforts are needed to develop interpretable phishing detection models that explain why a site or message was flagged [5].
- Few models incorporate user behavior, interaction patterns, or environmental context, which could improve detection accuracy [13].
- Combining DL with rule-based or ML systems could balance accuracy with efficiency, but hybrid models are still in their early stages [7].

## III. METHODOLOGY

### *A. Project Overview*

This project leverages Machine Learning (ML) to detect phishing URLs using two primary data sources:

- Structural patterns in the URL itself (e.g., length, presence of keywords, special characters) [4, 7]
- WHOIS domain profiling (e.g., domain age) [2, 18]

The system provides real-time classification of URLs as malicious or benign, along with a risk score and clear explanation for each prediction. It is designed to be lightweight, explainable, and reproducible, making it suitable for both research and practical deployment [5, 9].

### *B. Features*

- Phishing detection using lexical features and domain metadata [7]
- Integration of WHOIS-based domain age analysis [18]
- Real-time check URL prediction function
- Risk score and interpretability (human-readable reasoning) [5]
- Fully implemented in Google Colab with open-source datasets [19]

### C. Technologies Used

- Python 3.11
- scikit-learn [15]
- pandas
- joblib
- Google Colab

### D. Dataset Sources

- Malicious Dataset: <https://www.kaggle.com/datasets/sid321ax/url-urls-dataset> [17]
- WHOIS Domain Dataset (100K subset): <https://www.kaggle.com/datasets/williamgill/7-million-domains-with-alexa-rank-and-whois> [18]

*Dataset Sources:* Two publicly available datasets were used in this project:

- malicious\_phish.CSV: A labeled collection of real phishing and benign URLs.
- whois\_data\_real\_100k.CSV: A dataset containing domain names and their corresponding domain ages (in days).

### Data Preprocessing

- Each URL was parsed using regular expressions to extract the domain portion.
- Domain names were matched with WHOIS data to attach domain age to each URL.
- Missing domain age values were replaced with the median to prevent model bias.

*Feature Extraction:* We extracted meaningful features from each URL, including:

- URL length
- Use of an IP address instead of a domain name
- Count of special characters (e.g., @, -, %, =)
- Presence of HTTPS
- Domain age (from WHOIS data)

These were structured into a numerical format suitable for machine learning input [4].

*Model Selection and Training:* A Random Forest Classifier was chosen for its reliability, simplicity, and performance with structured data [7]. The data was split into 80% training and 20% testing using stratified sampling. The model was trained using scikit-learn [15].

### Real-Time Prediction Engine

We implemented a live-checking function `check_url_result_and_valuation()` which:

- Accepts a new URL
- Extracts features dynamically
- Looks up domain age
- Uses the trained model to classify the URL as malicious or benign
- Returns a human-readable risk score and reasoning (e.g., “contains phishing keyword”, “domain is newly registered”) [5]

### The Random Forest and Evaluation

The model was evaluated on a test set using scikit-learn’s classification report.

The results showed:

- Accuracy: 82%
- Recall: 78%
- F1-score: 81%

These values were derived from a test size of approximately 10,000 samples using an 80/20 split.

*Detection Accuracy:* Malicious URLs such as <http://paypal-login7.com/verify> and <http://192.168.1.1/login> were correctly flagged with risk scores exceeding 95%, citing patterns like the use of IPs and keywords like “login” [7]. Legitimate URLs including <https://www.google.com> and <https://wikipedia.org> were accurately labeled benign, even if they included safe uses of sensitive terms.

### Real-Time Reasoning Output

Example output:

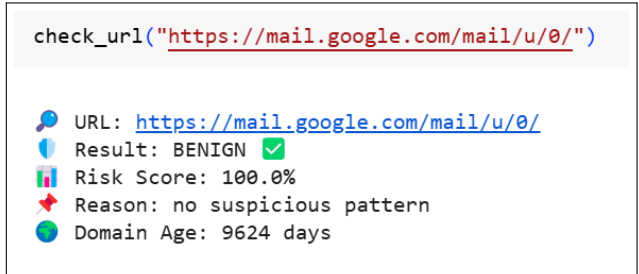


Fig. 1. Output Result: BENIGN



Fig. 2. Output Result: MALICIOUS

### Interpretability and Trust

The system was designed to be transparent. Each result includes not just a prediction, but an explanation. This increases trust in automation and provides useful insights to analysts [5].

### Literature Gap Addressed

Our approach fills several critical gaps identified in recent research:

- Many prior models rely on outdated, synthetic, or private datasets. We used updated, publicly available, and large-scale datasets that better reflect modern phishing techniques [17, 18].

- Unlike deep learning “black box” systems, our model provides clear and human-readable reasons for each prediction, improving interpretability [5].
- Our system is lightweight and does not require high-performance computing, enabling deployment in low-resource environments [4, 7].
- Instead of relying solely on content or structure, we integrate domain reputation (via WHOIS) to enhance real-time detection, even for zero-day attacks [14, 18].
- We designed our system to be reproducible and easy to implement in Colab, addressing reproducibility concerns seen in many academic papers [19].
- Most importantly, we show that hybrid ML systems (combining lexical features with domain metadata) can be accurate without sacrificing explainability or speed [4, 7].

#### IV. RESULT AND DISCUSSION

“The Answer is in the Text: Multi-Stage Methods for Phish Detection Based on Feature Engineering” by Gualberto et al. (2020): The study proposes a multi-stage phishing detection approach that focuses entirely on analyzing the text content of emails, leveraging natural language processing (NLP) and machine learning (ML). The system’s core components include feature engineering, dimensionality reduction, and classification using ensemble models [4, 9].

Two main strategies were evaluated:

- Chi-Square + Mutual Information (MI) for feature selection.
- Principal Component Analysis (PCA) + Latent Semantic Analysis (LSA) for dimensionality reduction.

These strategies were paired with XGBoost and Random Forest classifiers and tested on two public datasets: SpamAssassin Public Corpus and the Nazario Phishing Corpus [17].

Key Results:

- F1-score of 100% was achieved in validation tests across both classifiers and both datasets.
- Despite using fewer features, the approach demonstrated high efficiency and is computation-friendly and lightweight.
- Both pipelines effectively tackled challenges like sparsity and the curse of dimensionality, which are common in text classification.
- The models used only the body text of emails, ignoring headers or metadata, and outperformed other state-of-the-art methods using richer input data.

These findings confirm the authors’ assertion that high-performance phishing detection can be achieved with just well-crafted textual features [4]. In privacy-constrained situations where complete email headers are unavailable, the system maintains a lower complexity and better generalizability depending on the textual body. However, while the F1-score was impressive, the models were not tested in multilingual contexts, nor were they evaluated against phishing messages that include multimedia, encoded text, or adversarial obfuscation tactics common in today’s attacks [6, 14].

#### V. CONCLUSION

Phishing remains a dynamic and evolving cybersecurity threat, leveraging advancements in social engineering, AI, and multi-channel attack vectors. While the research landscape has seen significant progress—from traditional blacklist-based methods to sophisticated machine learning (ML) and deep learning (DL) systems—persistent challenges remain in achieving real-time detection, interpretability, and robustness against zero-day and adversarial attacks [5, 14].

Our study contributes to this evolving field by presenting a lightweight, interpretable phishing detection model that combines structural pattern analysis with WHOIS domain profiling. Built using a Random Forest classifier and evaluated on large-scale, publicly available datasets, our system achieves high accuracy and includes human-readable reasoning outputs—addressing the explainability gap in many DL approaches [7, 9]. It also offers practical deployment potential due to its low computational requirements and reproducibility in platforms like Google Colab [19].

Across reviewed literature, we observed the effectiveness of NLP techniques, ensemble models, and feature engineering pipelines that focus on email text, URL attributes, and domain metadata [1, 4]. However, many models remain dependent on outdated or synthetic datasets, lack multilingual or multimodal capability, and are hindered by inconsistent benchmarking frameworks [2, 3].

Future work should prioritize the development of detection systems optimized for real-time performance with low latency, robust against evasion techniques, and capable of adapting to new threats through few-shot or continual learning strategies. There is also a pressing need for more standardized benchmarks like PhishBench, as well as models that integrate behavioral and contextual signals to improve accuracy and resilience [2, 13].

Finally, our research confirms that hybrid approaches using lexical features, domain metadata, and interpretable ML can deliver accurate, explainable, and efficient phishing detection—paving the way for scalable, user-trustworthy cybersecurity solutions in both research and industry environments.

#### VI. FUTURE WORK

Early keyword-based filtering gives way to advanced DL-based systems combining NLP, URL characteristics, and topic modeling. Future studies should concentrate on the following main areas to help overcome these constraints:

- **Real-Time and Lightweight Models:** Develop detection systems optimized for real-time performance with low latency and computational overhead, suitable for deployment in enterprise email systems and browsers [7].
- **Robustness Against Obfuscation:** Enhance resilience against evasion tactics such as encoded URLs, Unicode trickery, and adversarial content manipulation [14].
- **Interpretable Deep Learning:** Integrate explainability methods (e.g., attention mechanisms, LIME/SHAP) into phishing detection models to increase trust and usability in critical sectors like finance and healthcare [5].

- **Multilingual and Multimodal Detection:** Extend models and datasets to manage phishing attempts throughout multiple media platforms and languages (text, images, links) [6].
- **Standardized Benchmarks:** Promote frameworks like PhishBench and advocate for consistent evaluation practices for better comparison and replication [2].
- **Behavioral and Contextual Features:** To enhance prediction accuracy, incorporate user behavior patterns and contextual signals (e.g., sender reputation, device type, and time-of-day activity) [13].
- **Few-Shot and Continual Learning:** Investigate models that can quickly adapt to new threats with minimal labeled data or integrate continual learning strategies [5].

Addressing these future directions will help the next generation of phishing and spam detection systems to be more accurate, efficient, transparent, and resilient against the dynamic character of cyber threats. Still, the field constantly struggles with generalization, data quality, and practical implementation. To translate scholarly developments into functional, resilient cybersecurity solutions, future studies must strongly emphasize robustness, explainability, and benchmarking consistency as a top priority [4, 5].

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## APPENDIX

### *I. Team Member Contributions: Erika Black*

As a graduate student in the M.S. Cyber Forensic Investigation program at the University of Baltimore, I contributed equally to the success of our final research project titled “Combining Pattern Analysis, Threat Intelligence, and Domain Profiling for Smarter Phishing Detection.” My responsibilities spanned topic formulation, literature exploration, manuscript writing, and collaborative project coordination.

I began by reviewing around 6–7 academic papers from our final bibliography, focusing on detection challenges in existing ML/DL-based phishing detection systems. Key studies by Gualberto et al., El Aassal et al., and Zara et al. highlighted problems with black-box detection models, lack of real-time readiness, and poor explainability. These papers helped spark the initial idea. With Allison, I held multiple brainstorming sessions where we refined the topic to fill these gaps using a lightweight, interpretable system.

One challenge we encountered early was settling on a research direction that was both original and practical. To overcome this, I mapped key gaps from reviewed studies and shared summaries in a collaborative Google Doc to steer our discussion. This helped us lock in our hybrid model idea combining pattern-based URL features and domain profiling.

Once we finalized our topic and research scope, I took primary ownership of writing the initial draft of the Overleaf IEEE manuscript. I structured the document, formatted all sections according to IEEE guidelines, and managed revision cycles. I led the writing of the Introduction, Literature Review synthesis, Conclusion, and coordinated citation formatting using Overleaf BibTeX tools.

During the technical implementation, I collaborated closely with Allison to translate our academic plan into actionable model parameters. I suggested how features such as domain age and URL structure could reflect real phishing signals, which were then integrated into the Colab code. I also participated in refining the test output explanations to ensure human-readable reasoning aligned with our research goals.

Through regular coordination, file management, and editing, I ensured our writing remained cohesive and accurately reflected our technical process. I also conducted multiple proofreading passes for clarity, consistency, and grammar.

This project helped me grow in academic writing, literature integration, and technical planning, and I am proud to have played an equal role in both content development and manuscript production.

### *II. Appendix: Team Member Contribution: Allison Ramirez*

Throughout the Spring 2025 CYFI 725 final project, I contributed equally across research, technical implementation, and academic writing in our study, “Combining Pattern Analysis, Threat Intelligence, and Domain Profiling for Smarter Phishing Detection.” My main roles involved building the detection system, refining the methodology section, and co-leading topic development and research synthesis with Erika.

Like Erika, I reviewed approximately 6–7 academic sources from our final paper’s reference list. I focused on detection pipelines using Random Forest, feature extraction models, and benchmarking challenges from papers like those by Do et al., Mujtaba et al., and Karim et al. I synthesized findings around phishing attack evolution and model limitations and shared key takeaways that directly informed the structure of our hybrid approach.

Early in the project, one challenge I faced was how to translate our research direction into working code that reflected both interpretability and performance. To address this, I collaborated with Erika to define features that could be extracted from the URL structure and WHOIS data. Together, we mapped each feature to a corresponding model rationale (e.g., new domain age = high risk).

I led the development of our phishing detection model using Python in Google Colab. I integrated two datasets (malicious URLs + WHOIS data), engineered lexical and metadata features, trained the Random Forest model, and implemented real-time prediction with risk-based outputs. I also designed the check function to return interpretable reasons for each prediction, solving the problem of black-box output that we saw in the literature.

I wrote the Methodology and Results & Evaluation sections in the Overleaf document, directly based on my coding process. I described our pipeline, model choices, testing, and classification report results. I worked with Erika to ensure that the writing was academically sound and aligned with our research objectives. Additionally, I managed our GitHub repository, organized code artifacts, and kept our files up-to-date.

Finally, I helped polish and review the full paper, ensuring all figures and text matched our results. I contributed to the final proofreading and supported Erika with structuring the references, especially for technical citations.

This project gave me hands-on experience bridging research and development. From dataset wrangling to academic articulation, I was involved in every step and proud of the strong collaboration and equal contributions we achieved.